

Luka Dolgov 414-708-5494, luka.dolgov@gmail.com

Computer Science and Math Student at Yale University, Intended Year of Graduation: 2028

GitHub: <https://github.com/LukaDolgov>

Professional Experience

Machine Learning Research Assistant – May 2025 - Present

- **Herb Scarf Fellowship/Tobin RA – Economics Department – May 2025 – July 2025**
 - Engineered a tunable machine learning model to generate a satellite-imaging-based wealth index using geospatial data.
 - Coordinated workflows with faculty and graduate researchers, contributing Python expertise to model development and to generate raster-based and statistical visualizations.
 - Leveraged Yale's High-Performance Computing Cluster for large-scale batch jobs and GPU-accelerated model training/testing.
 - Integrated the Clay Geospatial Foundation Model as an encoder and developed a custom decoder tailored for index creation.
 - Applied advanced ML engineering techniques, including parameter optimization and tensor manipulation, to improve model accuracy and scalability.
 - Supported team discussions by presenting intermediate results and proposing methodological improvements.

Skills and Projects

- **Projects – Personal interests**
 - Shakespearean Chatbot: Designed and trained a transformer-based chatbot on the complete works of Shakespeare to generate responses in Elizabethan style.
 - Coinbase Trading Bot: Built an automated trading system leveraging Coinbase's API for high-speed transaction monitoring, integrating LSTM models for price prediction and strategy optimization.
 - Video Generation with Google Gemini (Veo 3): Engineered a video generator that interfaces with the Gemini API to auto-generate creative prompts and produce vivid, AI-generated video outputs in short-form content.
 - *Older projects and ideas can be found on my GitHub above*
- **Technical Programming Skills**
 - Languages: Python, C, JavaScript
 - Frameworks & Libraries: PyTorch, TensorFlow, NumPy, Pandas, Rasterio, React
 - Databases & Backend: MongoDB, Node.js (JavaScript)
 - Tools: Git, High-Performance Computing (HPC), GPU utilization

Additional Information

- **Languages:** English-Native, Spanish-Advanced Level
- **For Fun:**
 - Violist, State-ranked player in Highschool, Yale Symphony Orchestra (2024-2025)
 - Chess player, 1650 online rating
 - Recreational Tennis and Ping Pong